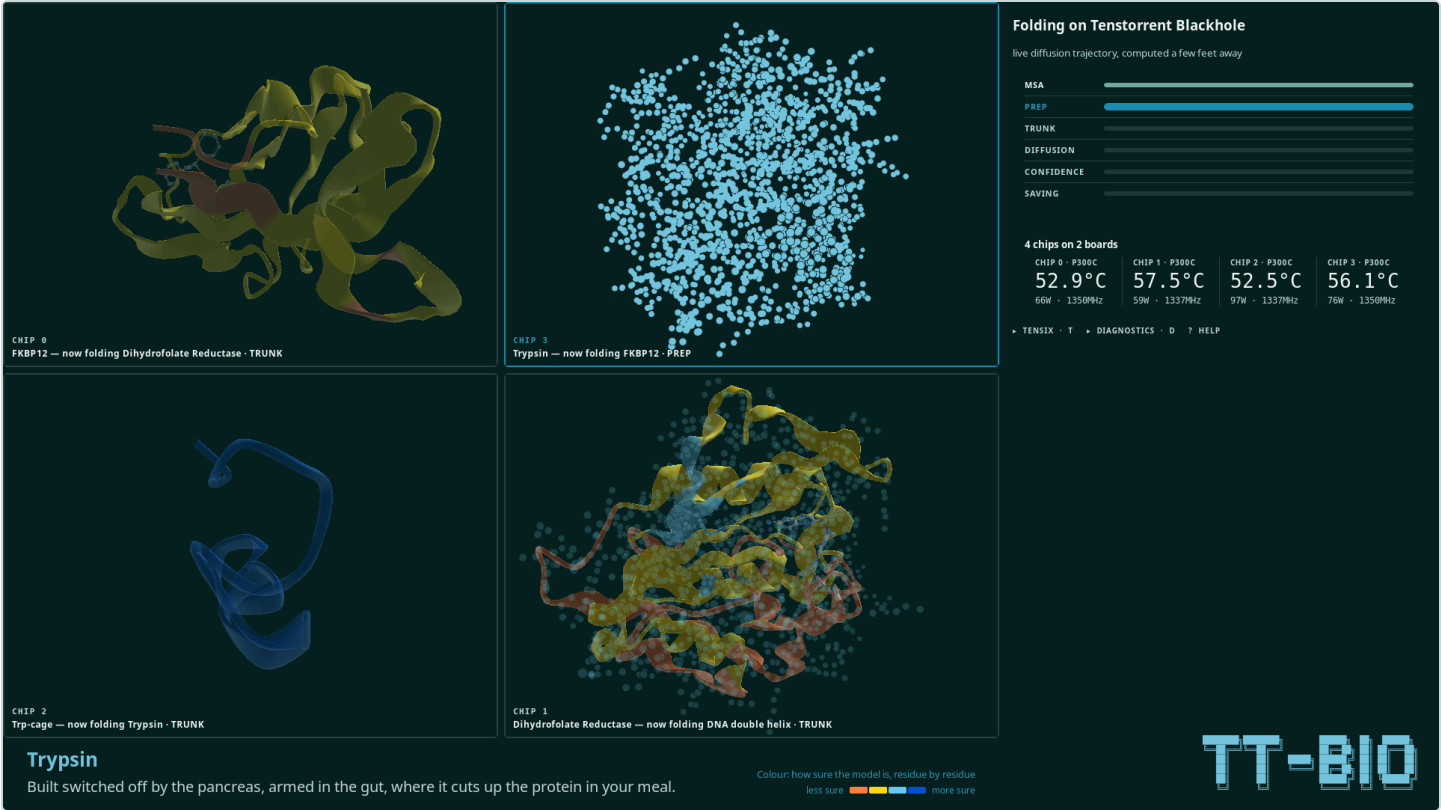


A protein is a chain of amino acids that only works once it has folded into a specific three-dimensional shape. Predicting that shape from the sequence alone was an open problem in biology for fifty years. This booth does it in front of you — [on the four Blackhole chips a few feet away](#), in seconds, with nothing pre-recorded.



The real application, on real silicon. Four Blackhole chips (two p300c boards), each folding a different molecule at its own point in the pipeline. Structures are drawn as a cartoon — helices as ribbons, sheets as arrows. The loose dots are the model's actual denoising trajectory, streamed one step at a time as it condenses into structure.

WHAT YOU ARE WATCHING

The model starts from **pure noise** — a cloud of atoms in random positions — and removes a little of that noise at a time, guided by what it has learned about how real proteins are built. Every dot on screen is a genuine intermediate step, sent from the chip as it happens.

Nothing here is an animation of a finished answer. When a fold has no coordinates yet, the booth holds the **previous real structure**, dimmed and labeled, rather than inventing something to show you.

COLOR = THE MODEL'S CONFIDENCE



Each residue is colored by **pLDDT** — the model scoring its own certainty. Blue regions it is sure about; orange it is guessing. An honest model tells you where it is weak, and this one is shown doing exactly that.

THE SIX STAGES ON THE PANEL

MSA	related sequences from other organisms — skipped here, which is why some targets come back orange
PREP	builds the model's input features
TRUNK	how every residue relates to every other. Makes no coordinates — so a big protein shows nothing new at first, and the booth keeps the previous structure up
DIFFUSION	the part you watch: noise becomes structure
CONFIDENCE	the model scores itself — where the colors come from
SAVING	writes the result out as mmCIF

The science is not ours. This booth exists to show other people's work running on Tenstorrent hardware. Folding is **tt-bio** (MIT) by Moritz Thüning; the demo around it is Apache-2.0. Structures are the model's own output, written as mmCIF and drawn here — no reference structure is ever shown in its place.

WHAT IT FOLDS

MOLECULE	SIZE	WARM
Trp-cage	20 res	4.5 s
DNA double helix	—	4.8 s
Transfer RNA	—	6.7 s
FKBP12 + binder	107 res	9.3 s
DHFR + methotrexate	187 res	13.8 s
Trypsin + benzamidine	223 res	17.4 s
Albumin the long one	585 res	84.4 s

Measured on this hardware, warm. Three fold a **bound ligand** alongside the protein — including DHFR with methotrexate, a chemotherapy drug, shown sitting in the pocket it blocks. Albumin is the long one: at 585 residues it is what a whole, real, drug-carrying protein costs.

Four chips, four proteins — not one protein four times faster.

A single fold runs on a single chip; that is tt-bio's documented limit, not something this demo works around. Four chips buy four folds at once, and a shorter wait for whatever you pick.

STARTING IT

```
# the normal way — double-click
# "tt-bio Protein Folding Demo"

# or, from a terminal:
/opt/tt-bio-demo/scripts/run-demo.sh
/opt/tt-bio-demo/scripts/run-demo.sh --solo
```

A booth with more than one chip **comes up in the 2x2 grid**; one chip comes up showing a single large protein. `--solo` forces the single view either way, `--quad` forces the grid, and **Q** toggles at runtime. The launcher starts **both** halves — the compute daemon and the UI — and tears the daemon down when it exits.

STOPPING IT

Ctrl + **Q**, or **Ctrl** + **C** in its terminal. One INFO line about a `tt-smi` sample killed by a signal is **normal** on a clean stop.

Signaling by pid does nothing — the launcher's trap cannot run until the UI returns. Signal the whole process group:

```
kill -INT -$(ps -o ppid= -p <pid> | tr -d ' ')
```

KEYS

? or F1	the help card, from any screen
Q	quad view — all four chips at once
T	Tensix activity — live core grids
D	diagnostics — the live protocol log
Esc	close the card, or either rail panel
Ctrl + G	easter egg: the Tenstorrent mark, computed on a chip
Ctrl + F	leave / return to fullscreen
Ctrl + A	restart the booth with affinity Q&A enabled (one-way; no-op if already on)
Ctrl + Q	quit
any other key or a tap anywhere	wake the booth and show what it folds

RUNNING IT UNATTENDED

For a booth that must survive all day, let systemd own the compute daemon so it comes back by itself:

```
systemctl --user enable --now tt-bio-demo
systemctl --user status tt-bio-demo
systemctl --user restart tt-bio-demo
```

It restarts **on failure only** — if you stop it deliberately it stays stopped. The UI is still launched separately, from the desktop entry.

BEFORE DOORS OPEN

- ☐ **Four chips present.** `tt-smi -s` lists four entries across two `board_id` s. Three means one did not come up.
- ☐ **Fold each target once.** This is the one that gets skipped. It warms the kernel cache so no visitor watches a cold compile.
- ☐ **Help card opens.** Press **?**. It is what a visitor reads when you are talking to someone else.

THE RIGHT-HAND RAIL

- Pipeline**
One row per stage — msa, prep, trunk, diffusion, confidence, saving. The bright row is running now; diffusion owns most of the bar. Clears itself after 20 s of silence, so a dead daemon cannot leave a bar frozen all day.
- Chips**
Temperature, power and clock per chip, from a `tt-smi` snapshot every two seconds — sampled on the UI's own thread, never through the daemon, so a wedged daemon still leaves the hardware readout live.
- Tensix activity — T**
One animated core grid per chip. Only the chip actually running the fold animates the work, and the header names it.
- Diagnostics — D**
Every protocol event as it arrives, with a plain-language line for each stage. Closes itself after five minutes of no input.
- Help card — ?**
Works from any screen without pausing the fold. Closes after a minute.

WHEN A VISITOR TAPS A PROTEIN

It gets folded **next** — never "now". The tap goes to the head of the queue and lands on the next chip to come free. Nothing already running is canceled, because tearing a fold down mid-device-operation is a known instability source.

Measured: **0.25–5.5 s** with all four chips busy (median ≈ 2 s). And most taps queue nothing at all — with seven targets on four chips the protein is usually already folding, so that cell simply takes focus. The booth acknowledges every tap immediately on a one-line notice.

Never run `tt-smi -r` on a shared machine. A chip missing from telemetry may have been deliberately quarantined by the daemon for temperature. Check `tt-smi -s` first.

Do the first fold before doors open — about 90 s per target. Every time on the front of this sheet is measured **warm**. On a machine that has never folded that target, the kernels compile first: measured **94.5 s** against 9.4 s warm. Fold each one you plan to show, once, while nobody is watching.

QUICK FIXES

A cell says TRUNK and looks frozen	Working as designed — trunk makes no coordinates. It is holding the last real structure and says so.
A chip missing from the chips panel	Check <code>tt-smi -s</code> . It may be quarantined for temperature on purpose.
Panels frozen, protein still moving	Telemetry is sampled separately from the daemon. Restart the daemon; the UI can stay up.
Everything stopped	Ctrl + Q , then relaunch from the desktop entry.

- ☐ **Weights on disk.** The venue is offline — nothing downloads at the booth. `doctor.sh` checks they are present *and* intact; `dpkg -l` only says a package is installed.
- ☐ **Fullscreen.** **Ctrl** + **F**. A multi-chip booth is already in the grid; **Q** toggles.
- ☐ **Leave it alone for a minute.** The attract loop should start folding on its own, unprompted.